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# EasyPep™ Magnetic MS Sample Prep Kit, 96 Reactions

Catalog Numbers A57867

Doc. Part No. 2162771 Pub. No. MAN0030132 Rev. B

**WARNING!** Read the Safety Data Sheets (SDSs) and follow the handling instructions. Wear appropriate protective eyewear, clothing, and gloves. SDSs are available from **thermofisher.com/support**.

## Product description

The Thermo Scientific™ EasyPep™ Magnetic MS Sample Prep Kit enables rapid and efficient processing of cultured mammalian cells, plasma, and tissues for proteomic mass spectrometry (MS) analysis. The kit is optimized to process protein samples from 15–100 µg with a high yield of peptides.

The kit contains pre-formulated buffers, MS-grade enzyme mix, magnetic beads for peptide clean-up, and an optimized protocol to generate MS-compatible peptide samples in less than 3 hours. Key features that reduce total sample preparation time include: addition of Universal Nuclease to reduce viscosity from nucleic acids without the need for sonication, a rapid "one pot" reduction/alkylation solution for cysteine modification (carbamidomethylation, +57.02), and a trypsin/Lys-C protease mix for more complete digestion.

The kit also includes magnetic clean-up beads and solutions to prepare peptide samples compatible with direct LC-MS analysis without the need to vacuum dry and reconstitute samples before injection. In addition, the peptide samples are compatible with isobaric tag (e.g., TMT™/TMTpro™ reagents) labeling and high pH reversed-phase fractionation.

The EasyPep™ Magnetic MS Sample Prep Kit, 96 Reactions can process samples manually or use automation instruments such as the Thermo Scientific™ KingFisher™ Apex Purification System.

## Contents

The EasyPep™ Magnetic MS Sample Prep Kit contains sufficient reagents for 96 preparations of 15–100 µg.

| EasyPep™ Magnetic MS Sample Prep Kit (Cat. No. A57867) | | |
|---|---|---|
| **Reagent** | **Amount** | **Storage** |
| Lysis Solution | 25 mL | Store at 4°C. |
| Reduction Solution | 7 mL | |
| Alkylation Solution | 7 mL | |
| Enzyme Reconstitution Solution | 1 mL | |
| Bind Solution | 4 × 26 mL | |
| Wash Solution | 2 × 26 mL | |
| Elution Solution | 11 mL | |
| Magnetic Clean-up Beads | 2.4 mL | Store at 4°C. Do not freeze. |
| Universal Nuclease | 25 kU | Store at 4°C.<br>Keep at -20°C for long-term storage. |
| Pierce™ Trypsin/Lys-C Protease Mix, MS Grade | 5 × 100 µg | |



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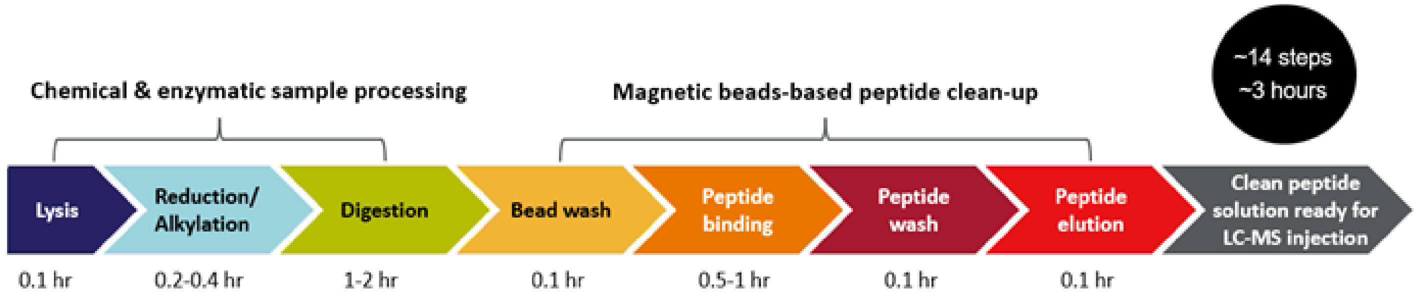
## Procedure summary




## Additional information

- Warm the Lysis Solution to room temperature before use. Store solutions at 4°C.
- Addition of phosphatase inhibitors to Lysis Solution (e.g., Halt™ Phosphatase Inhibitor Cocktail, Cat. No. 78420) is recommended before cell lysis for phosphopeptide enrichment.
- Addition of protease inhibitor cocktails containing ETDA to Lysis Solution is **NOT** recommended as these reagents inhibit Universal Nuclease and Trypsin/LysC Protease Mix activity.
- For long-term storage (>3 months), store Universal Nuclease and Trypsin/Lys-C Protease Mix at -20°C.
- After addition of Enzyme Reconstitution Solution, the Trypsin/Lys-C Protease Mix can be stored at 4°C for up to 1 month or -20°C for 1 year.
- Use of Magnetic Clean-Up Beads is required to remove contaminants and enzymes before LC-MS analysis.
- Do not dry or freeze the Magnetic Clean-Up Beads.

## Materials required but not supplied

- *(Optional)* Tissue homogenizer
- Heat block or thermal mixer
- Protein assay kit (e.g., Thermo Scientific™ Pierce™ BCA Protein Assay Kit, Cat. No. 23227)
- Low protein binding tubes (e.g., Thermo Scientific™ Low Protein Binding Collection Tubes, Cat. No. 90410)
- Magnetic stand (e.g., DynaMag™-2 Magnet, Cat. No. 12321D)
- Mass spectrometer with nano-flow liquid chromatography (LC) system

## Procedure for sample preparation

This procedure has been designed for 15–100 µg of protein sample input. Scale this protocol depending on the input protein sample range (Table 1). Initial protein samples in Lysis Solution should be ≥2–5 µg/µL to ensure the protein digest concentration is ≥0.06 µg/µL for the peptide binding and clean-up procedure. Sample amounts greater than 100 µg require a larger tube (e.g., 15-mL conical tube) for peptide cleanup.

### Extract or dilute protein sample

Rinse cultured cells or tissues 2–3 times with 1 × PBS to remove cell culture media or excess blood from tissues before adding Lysis Solution. Protein, cell, or tissue samples should be resuspended in Lysis Solution without additional buffers.

1. For cultured cells, add 25 µL of Lysis Solution and 1 µL of Universal Nuclease to a minimum of 1 × $10^6$ cells. Pipet up and down (with a P200 tip) 10–15 times until sample viscosity is reduced.

   **Note:** Centrifugation of cultured cell lysates is typically not required after aspiration using a pipette.

2. For tissue samples, add 25 µL of Lysis Solution (containing 1 µL Universal Nuclease) per 5 mg of tissue and disrupt with tissue homogenizer until sample is homogenized. Centrifuge tissue lysates at 16,000 × *g* for 10 minutes.

3. For purified proteins, serum, and plasma samples, dilute samples directly in Lysis Solution to 2–5 µg/µL.

   **Note:** For purified proteins and plasma samples, the addition of Universal Nuclease is **NOT** required.

4. Determine the protein concentration of the supernatant using established methods such as the Pierce™ BCA Protein Assay Kit (Cat. No. 23227) or Pierce™ Rapid Gold BCA Protein Assay Kit (Cat. No. A53226).

5. Adjust the final concentration of the protein sample to 2–5 µg/µL by adding the appropriate volume of Lysis Solution.

6. Transfer the appropriate amount of protein sample (see Table 1) into a 1.5-mL low protein binding tube, 2-mL 96-deep well plate, or 15-mL conical tube.

7. Proceed to Reduction, Alkylation, and Digestion using the reagent volumes indicated in Table 1.

**Table 1 Reagent volumes for different protein input ranges**

<table>
<tr><th rowspan="2">Protein range</th><th colspan="4">Protein digestion</th><th colspan="5">Peptide clean-up</th><th rowspan="2">Maximum reaction no.</th></tr>
<tr><th>Input Volume [1]</th><th>Reduction Solution</th><th>Alkylation Solution</th><th>Enzyme Solution</th><th>Magnetic Clean-up Beads</th><th>Bind Solution [2]</th><th>Bind Solution [3]</th><th>Wash Solution</th><th>Elution Solution</th></tr>
<tr><td>15–30 µg</td><td>7.5 µL</td><td>3.75 µL</td><td>3.75 µL</td><td>3.75 µL</td><td rowspan="3">20 µL</td><td rowspan="3">200 µL</td><td>230 µL</td><td rowspan="3">500 µL</td><td>30 µL</td><td rowspan="3">96</td></tr>
<tr><td>30–60 µg</td><td>15 µL</td><td>7.5 µL</td><td>7.5 µL</td><td>7.5 µL</td><td>460 µL</td><td>90 µL</td></tr>
<tr><td>60–100 µg</td><td>30 µL</td><td>15 µL</td><td>15 µL</td><td>7.5 µL</td><td>920 µL</td><td>150 µL</td></tr>
<tr><td>100–300 µg</td><td>75 µL</td><td>37.5 µL</td><td>37.5 µL</td><td>37.5 µL</td><td rowspan="3">200 µL</td><td rowspan="3">1 mL</td><td>2.3 mL</td><td rowspan="3">2.5 mL</td><td>300 µL</td><td>12</td></tr>
<tr><td>300–600 µg</td><td>150 µL</td><td>75 µL</td><td>75 µL</td><td>75 µL</td><td>4.6 mL</td><td>900 µL</td><td>10</td></tr>
<tr><td>600–1500 µg</td><td>300 µL</td><td>150 µL</td><td>150 µL</td><td>150 µL</td><td>9.2 mL</td><td>1.5 mL</td><td>5</td></tr>
</table>

[1] Volume of protein in Lysis Solution.
[2] Volume used to rinse Magnetic Clean-up Beads.
[3] Volume used for peptide binding.

### Reduce, alkylate and digest protein

1. Add the appropriate volume of Reduction Solution to the sample and briefly vortex.
2. Add the appropriate volume of Alkylation Solution to the sample and briefly vortex.
3. Incubate the sample at 95°C using a heat block for 5–10 minutes, or 50°C for 20–30 minutes to reduce and alkylate the protein sample.
4. After incubation, allow the sample to cool to room temperature. If necessary, briefly centrifuge the tube before digestion to collect protein sample at the bottom of the tube.
5. Add 150 µL of Enzyme Reconstitution Solution to 1 vial of Trypsin/Lys-C Protease Mix.
6. Add the appropriate volume of reconstituted enzyme solution to the reduced and alkylated protein sample solution.
7. Incubate with shaking at 37°C for 1–2 hours to digest the protein sample. For 60–100 µg of input protein, incubation with shaking at 37°C for 2–3 hours is recommended for adequate digestion of the protein sample.

   **Note:** At this point, the protein digest can be labeled with TMT™ reagents before peptide clean-up. If you are performing this protocol, proceed directly to "Label peptides with TMT™ reagent before peptide clean-up".
8. Proceed to manual or automated peptide clean-up using the reagent volumes indicated in Table 1.

## Manual peptide clean-up

1. Vortex Magnetic Clean-Up Beads to fully resuspend the beads before use.
2. Transfer Magnetic Clean-up Beads slurry to a tube or plate.
   - For 15–100 µg samples, transfer 20 µL Magnetic Clean-up Beads slurry to a low protein binding microcentrifuge tube or 96-well plate.
   - For 150–1500 µg samples, transfer 200 µL Magnetic Clean-up Beads slurry to a 15-mL conical tube.
3. Collect the beads using a magnetic stand. Remove and discard the storage buffer.
4. Remove the tube or plate from the magnetic stand, then add appropriate volume of Bind Solution to rinse the Magnetic Clean-up Beads.
5. Vortex briefly and collect the beads using a magnetic stand. Remove and discard the supernatant.
6. Add the appropriate volume of Bind Solution to the rinsed Magnetic Clean-up Beads, then transfer the reconstituted bead slurry to the protein digest solution.
7. Incubate the sample at room temperature for 30–60 minutes with mixing at 1,200 rpm on a thermomixer.
8. Collect the beads using a magnetic stand. Remove and discard the supernatant.
9. Remove the tube or plate from the magnetic stand, then add appropriate volume of Wash Solution, and vortex briefly.

   **Note:** For TMT-labeled samples, we recommend washing the resin twice using half the Wash Solution volumes in Table 1.
10. Collect the beads using a magnetic stand. Remove and discard the supernatant.
11. Add appropriate volume of Elution Solution, vortex briefly, and incubate for 3 minutes while mixing on a thermomixer at room temperature.
12. Collect the beads using a magnetic stand. Transfer the eluted peptide solution to a new tube or autosampler vial for LC-MS analysis.

    **Note:** We recommend centrifugation of eluted peptide samples at 10,000 × *g* for 2 minutes before transferring the eluted peptides to the autosampler vials to avoid bead carryover. Using C18 tips or trap columns is also recommended to ensure resin particles do not interfere with LC-MS analysis.

## Procedure for automated peptide clean-up

### Materials required but not supplied

- KingFisher™ Apex with 96 Deep Well Head (Cat. No. 5400930) or KingFisher™ Flex Purification System with 96 Deep-Well Head (Cat. No. 5400630)
- KingFisher™ 96 Deep-Well Plate, v-bottom, polypropylene (Cat. No. 95040450)
- 96 Deep-Well Tip Combs for KingFisher™ Flex Magnetic Particle Processor (Cat. No. 97002534)

### Guidelines for automated purification

- Load the KingFisher™ Apex instrument with the `EasyPep_Magnetic_Kit_15-30ug.kfx`, `EasyPep_Magnetic_Kit_30-60ug.kfx`, or `EasyPep_Magnetic_Kit_60-100ug.kfx` instrument script (see the *KingFisher™ Apex Purification System User Guide* for details on downloading the script).
- The following protocol is designed for use with the KingFisher™ Apex Magnetic Particle Processor. The protocol can be modified according to your needs directly on the instrument using the touch screen interface or by using the BindIx™ PC Software provided with the instrument.
- If fewer than 96 wells are used in a plate, fill the same wells in each plate. For example, if using wells A1 through A12, use these same wells in all plates.
- To ensure bead homogeneity, mix the vial by repeated inversion, gently vortexing, or rotating the platform before adding the beads to the Bead plate.
- Combine the Tip Comb with a Deep Well 96 Plate.
- See the instrument user manual for detailed instructions.

### Prepare instrument and set up plates

1. Dilute 20 µL Magnetic Clean-up Beads in 80 µL Protein Binding solution per sample.
2. Dilute the protein digest in Bind Solution to a final appropriate volume per sample.
3. Set up the plate as described in Table 2.

**Table 2 Pipetting instructions for the protein digest purification protocol using Thermo Scientific™ Microtiter DW 96 Plates**

| Plate # | Plate name | Content | Volume | Time | Mix speed |
|---|---|---|---|---|---|
| 1 | Tip plate | — | — | — | — |
| 2 | Bead plate | 20 µL Magnetic Clean-up Beads in 80 µL Bind solution | 100 µL | — | — |
| 3 | Bead wash plate | Bind solution | 200 µL | 1 minute | Medium |
| 4 | Sampe plate | Protein digest in Bind solution | 250 µL (15–30 µg protein digest),<br>500 µL (30–60 µg protein digest),<br>or 1000 µL (60–100 µg protein digest) | 40 minutes | Medium |
| 5 | Wash plate | Wash solution | 500 µL | 2 minutes | Medium |
| 6 | Elution plate | Elution solution | 50 µL–150 µL | 3 minutes | Medium |

### Perform automated purification

1. On the touchscreen interface, choose the protocol and select the Start button. See the *KingFisher™ Apex Purification System User Guide* for detailed information.
2. Open the door of the instrument.
3. Load plates into the instrument as directed by the touch screen display, placing each plate in the same orientation. Confirm each action on the touch screen.
4. After sample processing, remove the plates as directed by the touchscreen display.

## (Optional) Label protein digest with TMT™ reagent

Tandem Mass Tag™ (TMT™) is used in isobaric labeling as a method to quantify relative differences in protein samples. TMT™ labeling can be performed either immediately after protein digestion (i.e., before peptide clean-up) or after peptide clean-up. Labeling peptides with TMT™ reagents after clean up allows for measuring and normalizing peptide samples for equal mixing.

- For 1–10 µg digests, use at least 1:16, w:w, sample to TMT™ reagent, and 1:20, w:w, sample to TMTpro™ reagent.
- For 10–100 µg, use 1:4 or 1:8, w:w, sample to TMT™ reagent and 1:5, or 1:10, w:w, sample to TMTpro™ reagent.

The procedures below are for labeling 10–100 µg of individual samples.

### Label peptides with TMT™ reagent before peptide clean-up

1. Dissolve TMT™ reagent and TMTpro™ reagent in 100% acetonitrile according to instructions for the TMT™/TMTpro™ reagent.
2. Add an appropriate volume of dissolved TMT™/TMTpro™ reagent to each protein digest sample according to the recommended sample-to-tag ratio.
3. Incubate for 30–60 minutes at room temperature.
4. Add 5 µL of 5% hydroxylamine, 20% formic acid solution to each reaction to quench, then incubate for 5 minutes at room temperature.
5. Combine equal amounts of labeled protein digest into a microcentrifuge or 15 mL conical tube.
6. Proceed to to manual peptide clean-up protocol using the recommended volumes in Table 1 based on the total amount of combined protein input.

### Label peptides with TMT™ reagent after peptide clean-up

1. Dry the peptide sample using a vacuum centrifuge and resuspend the sample in an appropriate volume of 100 mM TEAB, pH 8.5 or HEPES, pH 8. Verify pH using pH paper.
2. Dissolve TMT™ reagent and TMTpro™ reagent in 100% acetonitrile according to instructions for the TMT™/TMTpro™ reagent.
3. Add an appropriate volume of dissolved TMT™/TMTpro™ reagent to each peptide sample according to the recommended sample-to-tag ratio.
4. Incubate for 30–60 minutes at room temperature.
5. Add 5 µL of 5% hydroxylamine solution to each reaction to quench, then incubate for 5 minutes at room temperature.
6. Combine equal amounts of each labeled sample into 1 tube.
7. Acidify the sample by adding 5% TFA until pH < 3. Verify pH using pH paper.
8. Desalt combined peptide samples using Pierce™ Peptide Desalting Spin Columns, Cat. No. 89852), stage tips, or equivalent.

## Troubleshooting

| Observation | Possible cause | Recommended action |
|---|---|---|
| High viscosity sample after lysis. | Universal Nuclease was not added. | Add 1 µL of Universal Nuclease per 100 µL of lysis buffer. |
| | Protease inhibitor cocktail with EDTA used. | Do not add protease inhibitor cocktails containing EDTA. |
| Incomplete digestion. | Low enzyme activity. | Store enzymes at 4°C for 1 month or -20°C for long-term stability. |
| | | Cool samples after reduction/alkylation to room temperature before the addition of the Trypsin/Lys-C protease mix. |
| | Insufficient digestion time. | Increase digestion time to 3 hours with shaking. |
| | Protease inhibitor cocktail used. | Do not add protease inhibitor cocktails. |
| Low peptide yield | Insufficient protein input. | Increase the amount of protein sample. |
| | Protein concentration too dilute. | Verify protein concentration is 2–5 µg/µL. |
| | Protein digest not sufficiently diluted in Bind Solution. | Protein digests must be diluted in Bind Solution by >90% by volume before peptide clean-up. |
| | Insufficient beads used. | Use a minimum of 20 µL of beads per 15–100 µg of protein digest. |
| Over-alkylation | Alkylation occurred for too long. | Alkylate at 95°C for 5–10 minutes or 50°C for 20–30 minutes. |
| Overestimation of peptide yield using peptide assays | Incomplete removal of contaminants during peptide clean-up. | Centrifuge eluted peptide samples at 10,000 × *g* for 2 minutes before LC-MS analysis. |
| | | Use C18 tips or trap columns to remove beads. |
| Beads present in elution sample | Beads present in elution sample. | Centrifuge eluted peptide samples at 10,000 × *g* for 2 minutes before LC-MS analysis. |
| | | Use C18 tips or trap columns to remove beads. |

## Related products

| Product | Cat. No. |
|---|---|
| EasyPep™ 96 Micro MS Sample Prep Kit | A57864 |
| EasyPep™ Mini MS Sample Prep Kit | A40006 |
| EasyPep™ Maxi MS Sample Prep Kit | A45734 |
| EasyPep™ 96 MS Sample Prep Kit | A45733 |
| EasyPep™ Lysis Buffer | A45735 |
| Pierce™ BCA Protein Assay Kit | 23227 |
| Pierce™ Rapid Gold BCA Protein Assay Kit | A53226 |
| Pierce™ Quantitative Colorimetric Peptide Assay Kit | 23275 |
| Pierce™ Quantitative Fluorometric Peptide Assay | 23290 |
| Pierce™ Trypsin/Lys-C Protease Mix, MS Grade | A40007 |

## References


- Waas, Matthew, *et al.* SP2: Rapid and automatable contaminant removal from peptide samples for proteomic analyses. *Journal of Proteome Research* 18.4 (2019): 1644–1656.


## Limited product warranty

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For descriptions of symbols on product labels or product documents, go to thermofisher.com/symbols-definition.

**Revision history:** Pub. No. MAN0030132 B

| Revision | Date | Description |
|---|---|---|
| B | 16 July 2025 | The dilution volume in the "Extract or dilute protein sample" section was corrected. |
| A.0 | 7 November 2023 | New document for EasyPep™ Magnetic MS Sample Prep Kit. |